Precision Laser Systems

Light engineered for quantum precision.

Our Precision Laser Systems are a complete family of narrow-linewidth laser sources, controllers, and supporting instruments built for quantum science, atomic physics, spectroscopy, and photonics. The range spans external cavity diode lasers covering a broad wavelength span and built to order for your experiment, ready-to-use plug-and-play diode lasers at popular fixed wavelengths, and the electronics needed to drive and stabilize them. Every laser is available in a free-space or fiber-coupled version, so it can be integrated directly into an optical setup or delivered cleanly to a remote experiment. Designed and tested in-house, each system combines stable, reliable performance with a compact footprint and straightforward day-to-day operation. Contact us to discuss your wavelength, power, and integration requirements, and we will configure a system around them.

Tunable External Cavity Diode Lasers (ECDL)

Tunable, narrow linewidth laser source, designed around your experiment.

Emission wavelength: 400 nm to 1300 nm

Optical power: 10 mW to 100 mW

Output: Free-space or fiber-coupled

Applications

- Atomic & molecular spectroscopy
- Laser cooling & trapping
- Quantum sensing & research

Contact us for full specifications and configuration options.

ApeX-Pro

Tunable laser system for the rubidium D1 and D2 transitions, based on our ECDL platform.

Emission wavelength range: 780 nm to 795 nm

Optical power: > 80 MW

Linewidth: < 1 MHz

Mode-hop-free tuning: > 25 GHz

Output: Free-space or fiber-coupled version available

Applications

- Rubidium spectroscopy
- Magneto-optical traps & laser cooling
- Atomic clocks & quantum sensing

Contact us for full specifications and configuration options.

Turnkey Diode Lasers

Fixed-wavelength diode lasers, supplied aligned and ready to operate.

Emission wavelength: 405 nm, 532 nm, 650 nm, 1550 nm

Optical power: 5 mW to 50 mW

Output: Free-space or fiber-coupled

Applications

- Fluorescence & imaging
- SPDC, QKD
- Telecom, photonics & teaching labs

Contact us for full specifications and configuration options.

ApeX-Ctrl

Laser controller combining a low-noise current driver, temperature controller, and piezo driver in one unit.

Current driver: Up to 2A, noise < 10 µA/√Hz

Temperature stability: 2 mK

Piezo driver: 0 to 100 V

Interfaces: USB

Applications

- ECDL & ApeX-Pro operation
- Frequency scanning & locking

Contact us for full specifications and configuration options. Customisation available.

Accessories

The following modules and instruments extend our laser systems into complete, frequency-stabilized setups. Each can be ordered with a new laser system or added to an existing one.

Digital Locking Module

Optional digital locking module for ApeX-Ctrl, for frequency stabilization to an atomic or cavity reference.

Control channels: Fast current feedback and piezo feedback

Features: Automatic relock, digital-processing, software-configurable

Fits: ApeX-Ctrl, no external servo needed

Configuration: Optional add-on module for ApeX-Ctrl, not sold separately

Applications

- Atomic transition locking

- Cavity locking
- Long-duration experiments

Contact us for full specifications and configuration options.

Compact Spectrometer

Compact spectrometer for verifying and monitoring laser emission wavelengths.

Precision: 0.1 nm to 0.05 nm

Wavelength: 400 nm to 1100 nm

Input: Free-space or fiber-coupled

Interface: USB

Applications

- Wavelength verification
- Laser wavelength tuning
- Optical testing

Contact us for full specifications and configuration options.

Quantum Memory

Store the light. Preserve quantum states.

Our Quantum Memory captures, stores, and retrieves quantum states of light on demand, a capability at the core of scalable quantum communication and computing. Built around a precision-engineered atomic ensemble, the system maps incoming photons onto long-lived atomic spin states and releases them when needed, preserving the delicate quantum information they carry. By holding and synchronizing photons across separate nodes, the Quantum Memory makes it possible to build quantum repeaters that extend entanglement over long distances, overcoming the losses that limit direct photon transmission. Engineered for low size, weight, and power, it brings a laboratory-grade capability into a 19-inch rack compatible, compact, field deployable package ready for integration into next-generation quantum hardware.

Key Highlights: Storage time up to 100 μs | Retrieval efficiency [> 25 %] | Fidelity [> 90 %]

Applications: Quantum Repeaters | Quantum Networking | Photonic Quantum Computing

Full technical specifications are available on request. Contact us to learn more about the Quantum Memory and how it can support your application.

EPS based QKD

Guaranteed security by the laws of physics.

Our Entangled Photon Source (EPS) based Quantum Key Distribution system delivers encryption keys whose security relies on quantum mechanics phenomenon rather than computational complexity. The EPS generates pairs of entangled photons and distributes one photon of each pair to two remote users. Because the measurement results of entangled photons are intrinsically correlated, both parties establish an identical secret key, while any attempt at interception disturbs the entanglement and is immediately revealed. Since the key is created by the measurements themselves, the source does not need to be trusted, adding a further layer of protection. Compatible with existing optical fiber and free-space links, the system provides future-proof security that remains resilient against attacks from both classical and quantum computers.

Applications: Secure Communications | Government & Defense | Finance & Critical Infrastructure

Full technical specifications are available on request. Contact us to learn more about the EPS based QKD and how it can support your application.

Quantum Repeater

Extending the reach of the quantum world.

Our Quantum Repeater solution brings together entangled photon sources, quantum memories, and entanglement swapping into an integrated platform for extending quantum links beyond the distance limits of direct fiber transmission. By dividing long distances into shorter segments and linking them through entanglement swapping, the repeater enables users to share quantum states securely and reliably across distances where photon loss would otherwise make transmission impossible, forming a key building block of the emerging quantum internet. This

Quantum Repeater is a practical choice for organizations that handle sensitive data, including government agencies, financial institutions, and operators of critical infrastructure, who need long-term protection for their most important communications over long distances.

Key Highlights: Compatible with standard telecom fiber

Applications: Long-Distance QKD | Quantum Internet | Long-Haul Secure Networks

Full technical specifications are available on request. Contact us to learn more about the Quantum Network and how it can support your application.

Quantum Gravimeter

Measuring gravity with atoms.

Our Quantum Gravimeter measures absolute gravitational acceleration with exceptional precision and long-term stability using cold-atom interferometry. Clouds of laser-cooled atoms are released into free fall, where a sequence of laser pulses splits, redirects, and recombines their matter waves. The resulting interference pattern reveals the local value of gravity, referenced directly to the fixed properties of atoms and the wavelength of light. With no mechanical moving parts and no need for periodic recalibration, the Quantum Gravimeter avoids the drift and wear that affect classical instruments, delivering reliable, repeatable measurements over extended periods.

Key Highlights: Absolute gravity measurement at μGal level.

Applications: Geophysical Survey | Resource Exploration | Infrastructure Monitoring | Navigation

Full technical specifications are available on request. Contact us to learn more about the Quantum Gravimeter and how it can support your application.

Rydberg Sensor

Sensing electromagnetic fields with atoms.

Our Rydberg Sensor uses atoms excited to highly energetic Rydberg states to detect and measure radio-frequency electric fields with outstanding sensitivity. In these states, atoms respond strongly to even weak electromagnetic fields, and this response is read out optically with lasers, converting the field directly into a precise measurement. Because the measurement depends on invariable atomic properties and fundamental constants, the sensor is inherently self-calibrated and traceable to SI units. A single compact atomic probe can operate across an exceptionally broad frequency range that would otherwise require many conventional antennas, and its all-dielectric construction minimizes disturbance to the field being measured. The result is a versatile, high-accuracy tool for RF metrology, detection, and communications.

Key Highlights: Broadband frequency coverage [Sub-kHz to 100s of GHz] | Small atomic probe/antenna

Applications: RF Test & Measurement | Communication | Spectrum Monitoring | Electronic warfare

Full technical specifications are available on request. Contact us to learn more about the Rydberg Sensor and how it can support your application.

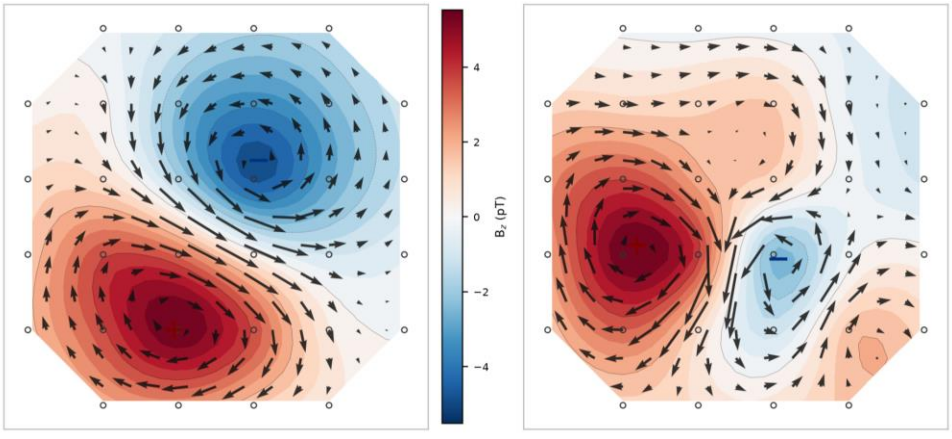
Quantum Biosensor

Magnetocardiograph

Our Magnetocardiography (MCG) platform captures the ultra-weak magnetic fields naturally generated by the heart, offering a new window into cardiac electrical activity. Powered by highly sensitive atomic magnetometers, the system provides a non-invasive, radiation-free and contrast-free view of cardiac electrophysiology. Because magnetic signals are less affected by surrounding tissues than conventional electrical measurements, MCG can reveal subtle spatial and temporal patterns in cardiac activity with greater clarity. Advanced signal processing transforms these signals into intuitive cardiac waveforms, magnetic field maps and clinically meaningful visualizations. Designed for speed, the platform rapidly captures and analyzes cardiac magnetic activity to support earlier identification of potential abnormalities and faster decision-making in time-sensitive and emergency care environments. Its compact sensing architecture simplifies MCG acquisition, while intelligent reconstruction techniques generate richer cardiac representations from a reduced sensor array. By combining advanced magnetic sensing, computational modeling and intelligent data analytics in a single workflow, the platform brings sophisticated cardiac assessment closer to practical clinical use. Built for both cardiovascular research and clinical decision support, our MCG platform is designed to make advanced cardiac insights faster, simpler and more accessible.

Key Highlights: Contact-free measurement | No electrodes, gel or skin preparation | Current map of the heart at each millisecond.

Applications: Cardiac Diagnostics | Acute and Emergency Care | Preventive Screening | Clinical Research



Healthy Diseased

Fig: Magnetic field map at the R-peak between a healthy and diseased Patient.

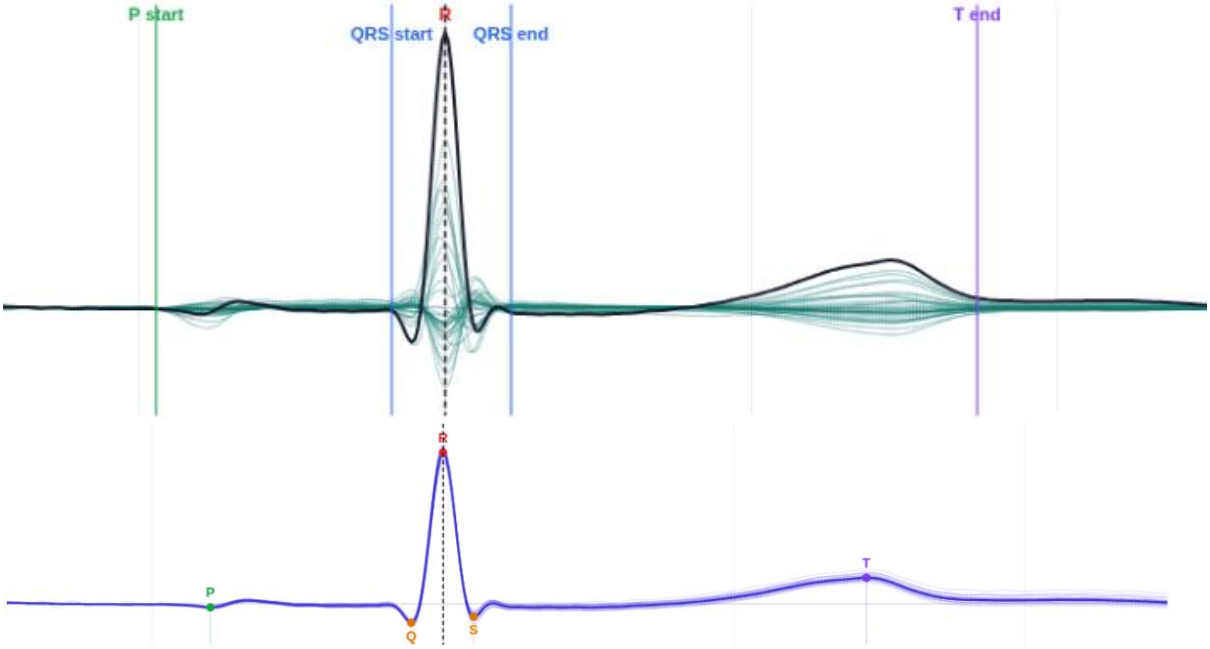


Fig: Butterfly Plot and averaged signal of the cardiac cycle.